

Hierarchical Reinforcement learning with AUVs

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Retireved from
<https://www.kongsberg.com/maritim e/products/marine-robotics/autonomous-underwater-vehicles/AUV-hugin/>

University of Oslo //
The Faculty of Mathematics and Natural Sciences //
Institute of Technology Systems //



Pollution Detection with AUVs

- Oceans cover 70.8% of Earth's surface
- Oceans are important ecosystem (e.g. produce significant amount of breathable air)

Danger of pollution,
but unreasonably large search area.

→ Search tasks for autonomous robots!



Retrieved from
<https://www.greenpeace.org/international/story/44923/oil-is-leaking-all-over-the-world/>

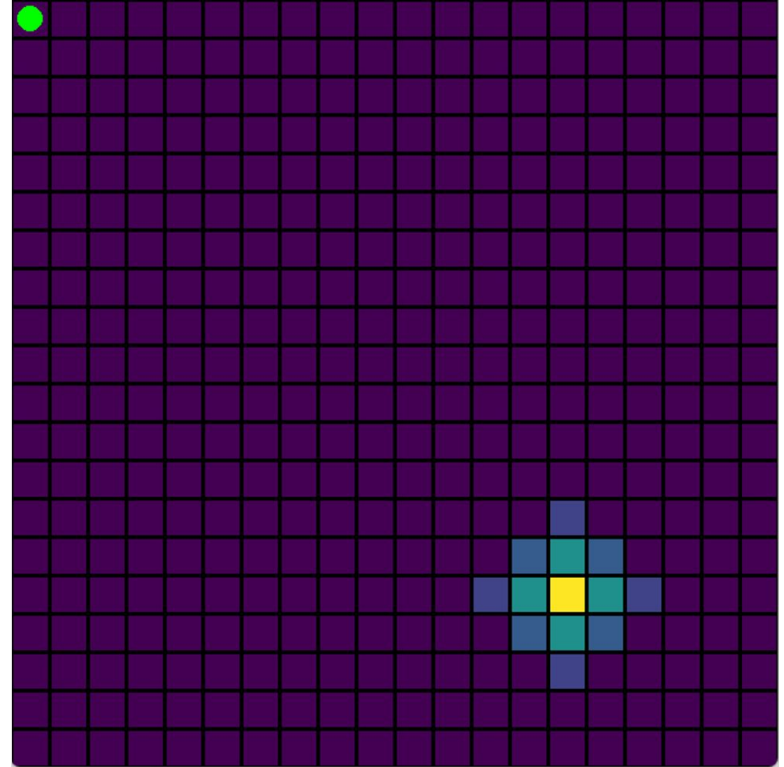
Environment Representation

- 2D Grid
 - AUV actor
 - Unknown pollution cloud

Abstraction over continuous 3D space of reality

Action: moving in 4 directions

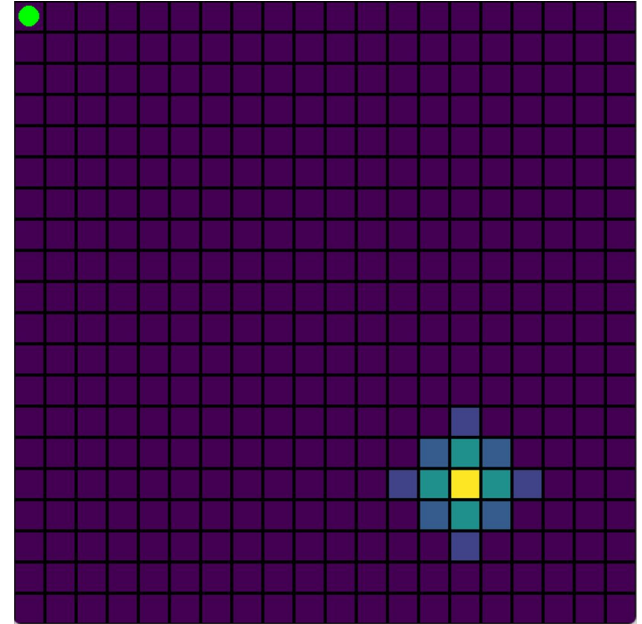
State: x, y



Problem Description

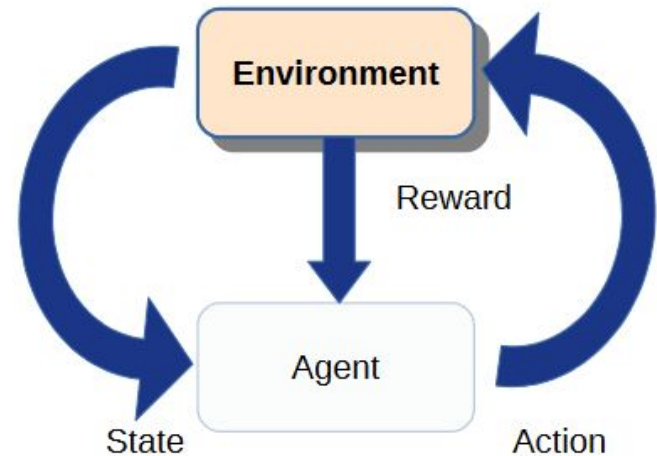
- Exploration Task: Find pollution
- Random, unknown pollution location

→ Find a location **without** information on where it may be, until you found it.



Reinforcement (Q-) learning Primer

- Find optimal policy mapping action to state in the environment
- Q-table documents Q-values together with states and actions
- Algorithm will learn optimal trajectory/sequence of state-action pairs via sensible reward function

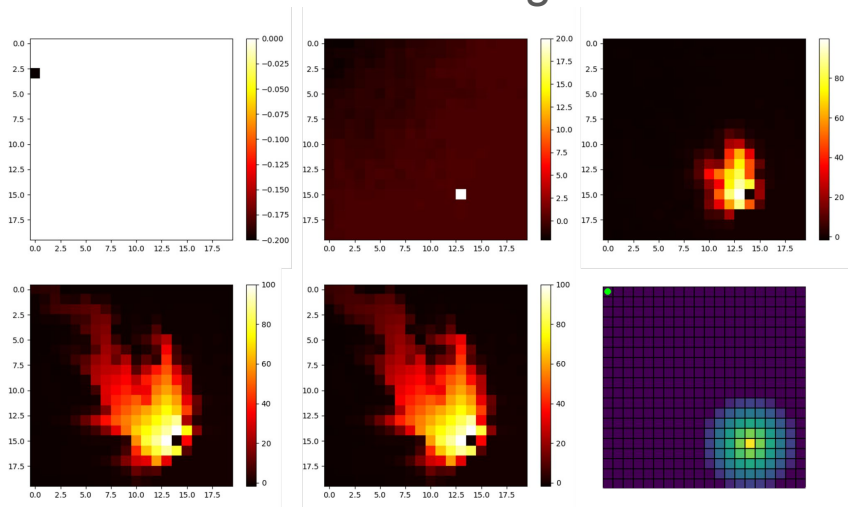


Borrowed from the lecture "Reinforcement Learning" in TEK5040

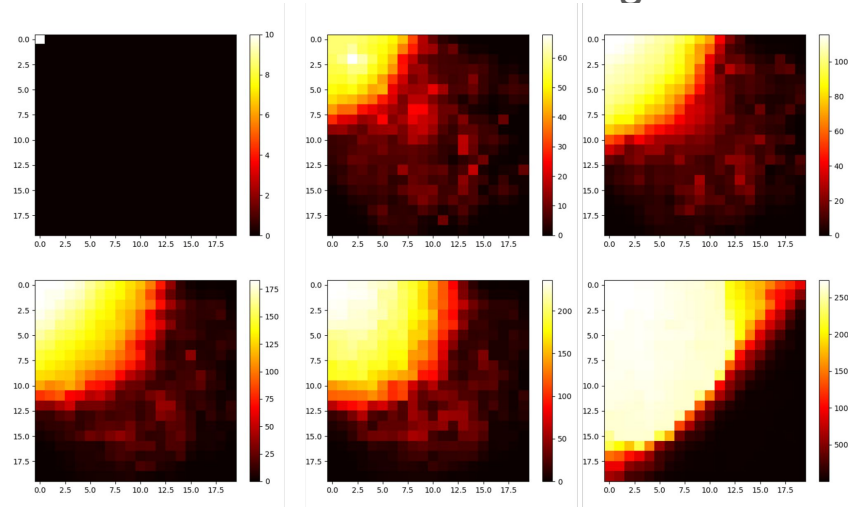
Methods

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a) \right)$$

Q-table with fixed setting:



Q-table with random setting

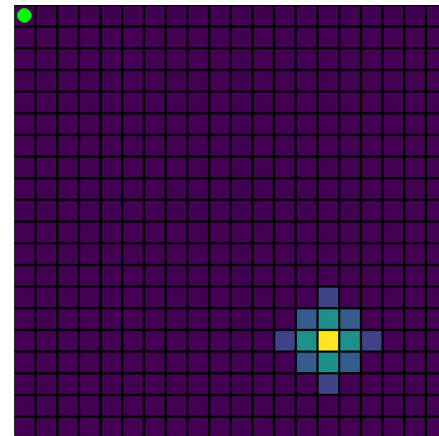
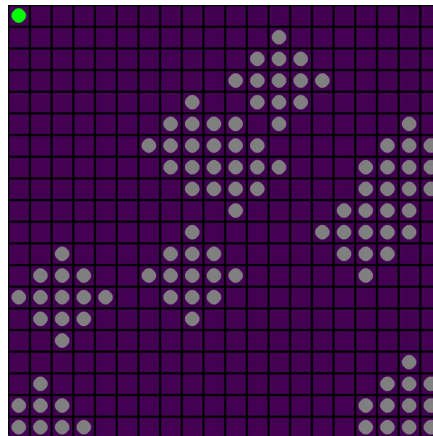


Methods

1. Hierarchical Reinforcement Learning
2. Multiple Goal Learning
3. All Step Learning

$$r = \frac{S_r}{n_{step}} \sum_{i=1}^{n_{poll}} 1$$

4. Best Experience Learning
5. Memory As Output Filter



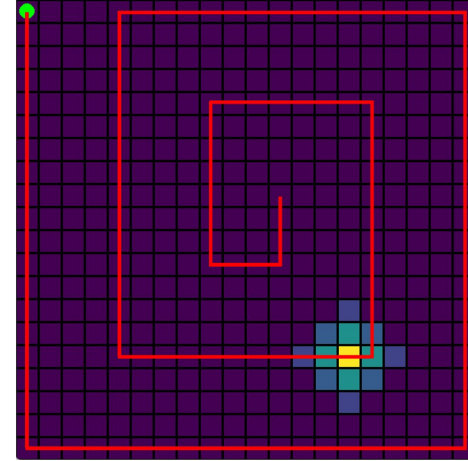
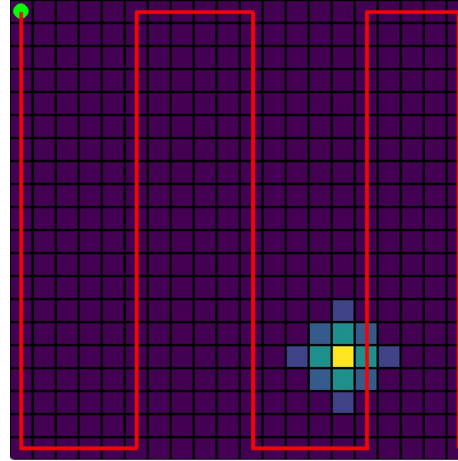
Methods

Evaluation:

What is the purpose of finding an RL-based strategy?

→ Comparison with manually developed patterns

→ Rating by duels scores and medians



Results

1000 Episodes: Score: HRL: **531**, Snake: 426, Tied: 42

Score: HRL: **505**, Spirale: 459, Tied: 35

Median: Snake: 41.0, Median Spirale: 52.0 **Mean HRL: 40.486, Median HRL: 38.0**

Maximum number of steps: 150

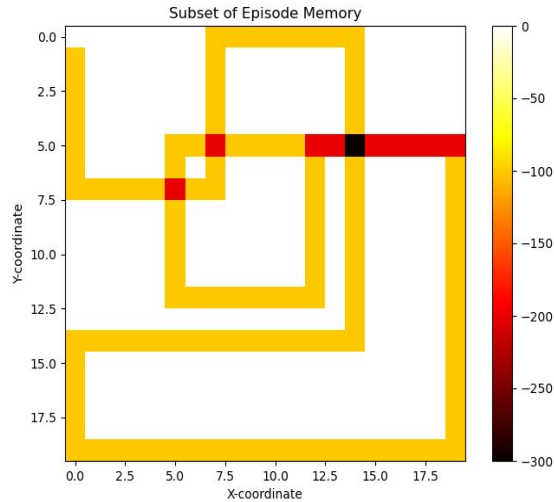
10000 Episodes: Score: HRL: **697**, Snake: 268, Tied: 34

Score: HRL: **611**, Spirale: 350, Tied: 38

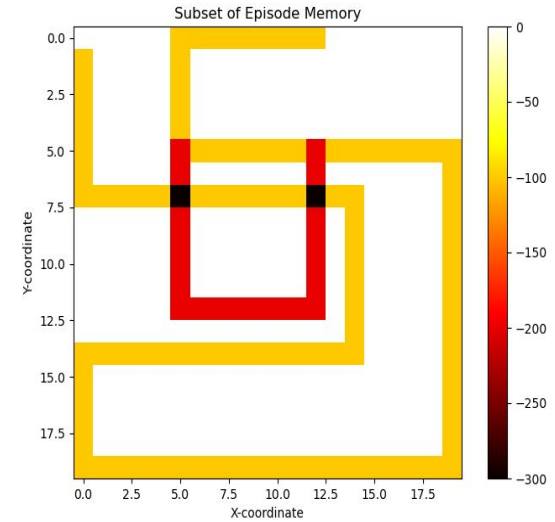
Median: Snake: 41.0, Median: Spirale: 52.0 **Mean: HRL: 49.663, Median: HRL: 32.0**

Maximum number of steps: 166

Results



1000 Episodes



10000 Episodes

Discussion

1. Hierarchical Reinforcement Learning stabilizes the Searching process and speeds up Learning clearly
2. Multiple Goal and All Step Learning is essential for overcoming the Randomness-problem and very successful
3. Memory as Output Filter for Q-Learning is inevitable
4. Best Learn needs further investigation

Future work

- More even pollution spawn distribution
- Expansion into other applications
- Deep Q-learning with memory as additional input
- Continuous, 3D environment with varying parameters

Thank you for your attention!



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